

VINAY

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PROFESSIONAL SUMMARY

Final-year ECE student at VIT Chennai (GPA 9.14) with experience in embedded firmware, IoT systems, FPGA design, and real-time communication protocols (UART/SPI/I2C/CAN/MQTT). Skilled in Embedded C/C++, Python, FreeRTOS, and hardware debugging. Seeking Embedded Software, Firmware roles.

TECHNICAL SKILLS

Programming & Scripting: C, C++, Embedded C, Python
Embedded & Real-Time Systems: FreeRTOS, Interrupt Handling, DMA, PWM, ADC/DAC, Sensor Integration
Communication Protocols: UART, SPI, I²C, CAN, MQTT
Embedded Platforms: ESP32, Arduino Uno, Jetson Nano .
Digital Design & Hardware: SystemVerilog, FPGA Design, RTL Design .
Tools & Debugging: Git, Oscilloscope, Logic Analyzer ,ModelSim, Intel Quartus Prime

EDUCATION

Vellore Institute of Technology <i>B.Tech – Electronics and Communication Engineering</i> Chennai, Tamil Nadu GPA: 9.14 / 10	2022 – 2026
St. Joseph's International School, Kota (Rajasthan) Senior Secondary (XII) 87%	2020
Modern Arya Public School, Faridabad (Haryana) Secondary (X) 83.4%	2018

INTERNSHIP EXPERIENCE

Mafkin Robotics Pvt. Ltd. (ECE Intern) Chennai, TN India Drive	Oct 2024 – Jan 2025
- Debugged and integrated embedded subsystems (microcontroller firmware, sensors, and communication interfaces) for robotics platforms, using structured root-cause analysis and oscilloscope/logic analyzer-based testing to cut issue-resolution time by ~25% and reduce integration-test failures by ~20%. - Collaborated with cross-functional hardware/software teams to integrate sensor modules and control logic, improving overall system response time and reliability.	

PROJECTS

AI Power Touch-Free Alert System (ESP32 + Jetson Nano) Github	Apr 2026
- Integrated ESP32-based sensor nodes with MQTT communication and OpenCV-based blink detection on Jetson Nano to deliver real-time patient alerts . - Designed low-latency (<2s) MQTT communication from ESP32 nodes to a centralized dashboard, with Telegram Bot API integration for automated emergency alerts. Stack : Python, OpenCV, MQTT, ESP32, Jetson Nano .	
Real-Time Avionics Primary Flight Display (C++ + Data Processing) Github	Sep 2025
- Built a real-time data acquisition and visualization pipeline for sensor-based flight parameters with noise-filtering algorithms. - Implemented UART-based serial data transmission and resolved latency/data-consistency issues to optimize end-to-end performance. Stack : C++, Python, UART, Sensor Data Processing	
Real-Time Lossless JPEG-LS Compression on FPGA	May 2025
- Implemented and validated a high-throughput FPGA image compression pipeline, fixing timing violations and clock-domain-crossing issues to reach stable 103 MHz operation (4x throughput gain), verified via ModelSim. Stack: SystemVerilog, Intel Quartus Prime, ModelSim.	

CERTIFICATIONS

Digital Design Training – Maven Silicon (MSUID: MS/V-DI/2024-25/583) Drive	Jun–Jul 2024
Verilog, RV32I RISC-V processor fundamentals, UVM verification basics, AHB interface design with hands-on lab exercises.	

LEADERSHIP & ACHIEVEMENTS

- Qualified GATE (ECE) and Solved 300+ Data Structures & Algorithms problems on LeetCode;
- Coordinated a finance awareness session as a member of Dream Merchants Club, managing event logistics and audience engagement.